

# Difference-based Sample Selection for Federated Graph Rationalization

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## Abstract

Graph rationalization methods aim to improve the explainability of Graph Neural Networks by identifying critical subgraphs (rationales) for task prediction. Motivated by increasing concerns over data privacy, federated graph rationalization has recently gained traction as a novel research area. However, in federated settings, data heterogeneity across clients exacerbates shortcut learning, where models rely on spurious and client-specific features rather than invariant causal rationales. Existing solutions, such as environment-aware data augmentation, suffer from low-quality environment representations. To address this, we propose *DiffGR*, a Difference-based sample selection strategy for federated Graph Rationalization. *DiffGR* selects samples where local and global models exhibit the highest prediction discrepancies, as these likely reflect strong shortcut reliance, enabling more accurate environment representations. Additionally, we introduce a mutual information (MI) inspired environment-conditioned data augmentation method that minimizes MI between environments and predictions while maximizing MI between rationales and predictions. Experiments on real-world and synthetic datasets demonstrate the effectiveness of *DiffGR* in improving rationale quality and model robustness in federated settings. Code is available at <https://github.com/yuelinan/Codes-of-DiffGR>.

## CCS Concepts

• Mathematics of computing → Graph algorithms.

## Keywords

Federated Learning, Graph Rationalization

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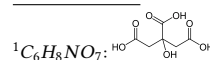
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## 1 Introduction

With the rapid development of Graph Neural Networks (GNNs), explainable GNNs have become an increasing focus [9, 11, 32, 37, 42], particularly graph rationalization methods that extract subgraphs as explainable rationales [23, 33, 47]. Commonly, a classic graph rationalization framework consists of two core components: (1) a *rationale extractor* that identifies critical subgraphs (i.e., rationales) from the original graph, and (2) a *task predictor* that performs tasks solely based on the extracted subgraphs. This framework aims to capture the most causally relevant nodes and edges while filtering out task-irrelevant or redundant graph components. For example, considering the task of predicting whether *Citric Acid* ( $C_6H_8NO_7$ )<sup>1</sup> is slightly water-soluble, the graph rationalization method identifies the carboxyl functional group ( $-COOH$ ) as the predictive rationale, where this subgraph exhibits strong hydrogen-bonding capability with water molecules.

Driven by growing demands for data privacy protection, federated graph rationalization [45] has emerged as a new research direction. Its basic training framework involves: Local training of rationalization models on clients → Server-side aggregation of extractor and predictor parameters from all clients → Distribution of the global model to clients for iterative training. However, data heterogeneity impedes further development of this framework [39, 45]. Specifically, differences in data collection methods across clients lead to significant variations in their data distributions (i.e., environments). This data heterogeneity exacerbates the “*shortcut learning*” problem [1, 37] in federated graph rationalization. Locally, models tend to capture client-specific shortcut features (i.e., spurious correlations that are statistically associated with labels but lack causal relationships) while ignoring truly invariant causal features (i.e., rationale features). Since these spurious correlations are highly sensitive to local data environments, aggregating conflicting shortcut features learned by different clients results in significantly degraded performance of the global model compared to centralized training scenarios.

Therefore, [46] propose an environment-aware data augmentation method. Recognizing that environment information is typically independent of prediction tasks, they employ the task-irrelevant complement subgraphs (non-rationales) to infer environment information. This approach clusters complement representations from all client samples to derive environment prototypes, enabling cross-environment data augmentation. However, this method suffers



from limited environment representation quality. Specifically, complement subgraphs may contain two types of interference: the environment information and task-irrelevant noise. Indiscriminate use of all samples for environment inference introduces substantial low signal-to-noise ratio data, ultimately compromising representation quality. This raises a critical question: *How to select the most representative samples to improve environment inference?*

To this end, in this paper, we propose a *Difference-based sample selection strategy for federated Graph Rationalization (DiffGR)*. Specifically, our key insight is that in federated learning, local models are more susceptible to shortcut features than the global model [39]. We quantify this phenomenon by computing the difference scores of the prediction performance between local and global models for each sample: higher scores indicate greater reliance on client-specific shortcuts, making their complement information more valuable for the environment inference. By selecting the top- $n\%$  samples with highest difference scores for clustering<sup>2</sup>, we can obtain more representative environment representations.

Furthermore, building upon these environment representations, in *DiffGR*, we propose a mutual information (MI) inspired environment conditioned data augmentation method. Specifically, from an information theoretic perspective, we first analyze the causal relationships among rationales, environments, and prediction results. Then, to enhance the extraction of environment-invariant rationales in the data augmentation, we simultaneously optimize two information theoretic objectives: (1) minimizing the MI between environments and prediction results while (2) maximizing the MI between rationales and results. Experiments over real-world benchmarks [15, 18] and various synthetic datasets [37] validate the effectiveness of our methods.

## 2 Related Work

### 2.1 Graph Rationalization

The rapid progress in Graph Neural Networks (GNNs) [8, 10, 25, 35] has spurred growing interest in explainability for graph classification tasks [2–5, 19]. Among emerging approaches, graph rationalization has gained prominence. However, as demonstrated by [1, 22, 24], these methods often exploit shortcut features in data rather than uncovering real predictive patterns. Consequently, recent research has prioritized the development of faithful rationalization techniques that mitigate shortcut learning. For example, DIR [37] proposed disentangling input graphs into rationale and non-rationale subgraphs, treating the latter as the environments to generate counterfactual samples. Subsequent work, such as GIL [19] and C2R [47], departed from DIR’s approach by inferring local environments through clustering non-rationale representations within mini-batches. Similarly, HSE [29] and EQuAD [41] leveraged environment inference to assign environment labels per sample. Alternative strategies have focused on architectural refinements to suppress shortcuts. DARE [44] introduced a self-guided framework that captures comprehensive input information via disentanglement, while GSAT [28] incorporated information bottleneck theory via a stochasticity-reduced attention mechanism. Further advancing this direction, DIVE [34] enforced subgraph diversity regularization to promote varied rationale discovery.

<sup>2</sup>Notably, when  $n=100$ , this strategy degenerates to the method in [46].

While rationalization methods have been widely investigated for centralized datasets, their application to federated learning (FL) scenarios remains largely unexplored. Addressing this gap, FedGR [45] introduced the first federated graph rationalization framework, which exploits discrepancies between global and local models in FL to develop difference-aware data augmentation strategies. This approach generated anti-shortcut samples for individual clients, significantly improving rationalization performance in distributed settings. Besides, EaDA [46] proposed an environment-aware data augmentation technique that leverages task-irrelevant complement subgraphs (non-rationales) to infer environmental information. By clustering complement representations across all client samples to derive environment representations, the method enabled effective cross-environment augmentation, further advancing the robustness of federated graph rationalization.

### 2.2 Federated Learning

Federated Learning (FL) has emerged as a pivotal paradigm for addressing critical data privacy and security challenges in distributed machine learning systems [26, 27, 36, 38, 40]. This decentralized approach enables multiple parties to collaboratively train models while keeping raw data localized, thereby preserving confidentiality and complying with increasingly stringent data protection regulations. Notable advances include the differentially private knowledge transfer framework by [30], which utilized actively selected public data to maintain model quality while preserving privacy. Further innovations in knowledge distillation were achieved by [31], which developed a selective sharing mechanism to optimize the transfer of accurate local predictions. Recent work has particularly focused on addressing the critical challenge of spurious correlations in FL systems. [6] introduced a fairness-aware FL framework that actively mitigates demographic biases during training. Complementing this approach, [39] proposed a novel augmentation method that incorporates causal attributes to systematically eliminate shortcut learning in federated environments. These developments collectively advance the robustness and reliability of FL systems for sensitive real-world applications.

## 3 Preliminaries

### 3.1 Problem Formulation

In this subsection, we present a formalization of graph rationalization in federated learning (FL) scenarios. Consider a federated system comprising  $N$  clients, denoted as  $\{C_1, C_2, \dots, C_N\}$ , each possessing a local graph dataset  $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_N\}$ . Notably, these clients exhibit inherent heterogeneity in their data distributions, reflecting the non-IID nature of federated environments.

For each client  $C_k$ , a graph-label pair is represented as  $(G_k, Y_k) \in \mathcal{D}_k$ , where  $G_k = (\mathcal{V}, \mathcal{T})$  with  $\mathcal{V}$  as the node set and  $\mathcal{T}$  as the edge set. The local graph rationalization task involves two key components:

(1) *Rationale Extraction*: A mask variable  $\mathbf{M}_k \in \mathbb{R}^{|\mathcal{V}|}$  and node representations  $\mathbf{H}_{G_k} \in \mathbb{R}^{|\mathcal{V}| \times d}$  are derived via a rationale extractor  $f_{s_k}(G_k)$ .

(2) *Task Prediction*: The rationale subgraph, obtained through element-wise multiplication  $\mathbf{M}_k \odot \mathbf{H}_{G_k}$ , is fed into a predictor  $f_{p_k}(\cdot)$  to generate the final output.

The objective is to jointly optimize the extractor  $f_{s_k}^*(\cdot)$  and predictor  $f_{p_k}^*(\cdot)$  by minimizing the cross-entropy loss  $\ell(\cdot)$  over  $\mathcal{D}_k$ :

$$f_{s_k}^*(\cdot), f_{p_k}^*(\cdot) = \arg \min_{f_{s_k}, f_{p_k}} \mathbb{E}_{(G_k, Y_k) \sim \mathcal{D}_k} [\ell(f_{p_k}(f_{s_k}(G_k)), Y_k)].$$

Over  $T$  communication rounds, the global goal is to aggregate client models into a global extractor  $f_s(\cdot)$  and predictor  $f_p(\cdot)$ :

$$\hat{\Theta}^s = \sum_{k=1}^N \frac{|\mathcal{D}_k|}{\sum_{j=1}^N |\mathcal{D}_j|} \Theta_k^s, \quad \hat{\Theta}^p = \sum_{k=1}^N \frac{|\mathcal{D}_k|}{\sum_{j=1}^N |\mathcal{D}_j|} \Theta_k^p, \quad (1)$$

where  $\hat{\Theta}^s$  and  $\hat{\Theta}^p$  denote the global parameters for  $f_s(\cdot)$  and  $f_p(\cdot)$ , respectively, while  $\Theta_k^s$  and  $\Theta_k^p$  represent the local parameters of client  $C_k$ .

### 3.2 Basic Training Framework of Federated Graph Rationalization

This subsection details the Basic Federated Graph Rationalization (Basic-FedGR) framework, comprising a rationale extractor and a task predictor.

**3.2.1 Rationale Extractor in Basic-FedGR.** For each client  $C_k$  with graph-label pair  $(G_k, Y_k) \in \mathcal{D}_k$ , the rationale extractor  $f_{s_k}(\cdot)$  operates as:

(1) *Node Encoding*: A GNN encoder  $\text{GNN}_m(\cdot)$  first maps each node in  $G_k$  to a  $d$ -dimensional vector.

(2) *Rationale Probability Prediction*: Then, the extractor predicts a node selection probability distribution:

$$\tilde{\mathbf{M}}_k = \text{softmax}(W_m(\text{GNN}_m(G_k))),$$

where  $W_m \in \mathbb{R}^{2 \times d}$  is a learnable weight matrix.

(3) *Differentiable Sampling*: Binary masks  $\mathbf{M}_k = \{m_k^i\}_{i=1}^{|\mathcal{V}|}$  are sampled using Gumbel-softmax [16] for differentiability:

$$m_k^i = \frac{\exp\left(\left(\log \tilde{m}_k^i + q_k^i\right) / \tau\right)}{\sum_t \exp\left(\left(\log \tilde{m}_k^t + q_k^t\right) / \tau\right)},$$

where  $m_k^i \in \{0, 1\}$ ,  $\tau$  is a temperature parameter,  $q_k^i = -\log(-\log(u_k^i))$ , and  $u_k^i \sim U(0, 1)$ .

(4) *Node Representation Extraction*: Finally, another GNN encoder  $\text{GNN}_G(\cdot)$  generates the node representations  $\mathbf{H}_{G_k}$  of the graph  $G_k$ . The rationale and complement (i.e., the non-rationale) representations are computed as  $\mathbf{M}_k \odot \mathbf{H}_{G_k}$  and  $(1 - \mathbf{M}_k) \odot \mathbf{H}_{G_k}$ , respectively.

**3.2.2 Predictor in Basic-FedGR.** The predictor  $f_{p_k}(\cdot)$  consists of a readout function and a classifier. Among them, the readout function aggregates rationale and complement subgraph representations to achieve derive the graph-level rationale  $\mathbf{h}_{r_k}$  and complement  $\mathbf{h}_{e_k}$  representations:

$$\begin{aligned} \mathbf{h}_{r_k} &= \text{READOUT}(\mathbf{M}_k \odot \mathbf{H}_{G_k}), \\ \mathbf{h}_{e_k} &= \text{READOUT}((1 - \mathbf{M}_k) \odot \mathbf{H}_{G_k}). \end{aligned}$$

Then, the classifier predicts labels using the rationale representations:

$$\hat{Y}_{r_k} = \Phi(\mathbf{h}_{r_k}), \quad \mathcal{L}_r^k = \mathbb{E}_{(G_k, Y_k) \sim \mathcal{D}_k} [\ell(\hat{Y}_{r_k}, Y_k)]. \quad (2)$$

**3.2.3 Training and Inference.** During training, to control rationale size, a sparsity loss [21] is applied on the probability  $\mathbf{M}_k$  of being selected as a rationale:

$$\mathcal{L}_{sp}^k = \left| \frac{1}{|\mathbf{M}_k|} \sum_{i=1}^{|\mathbf{M}_k|} m_k^i - \alpha \right|,$$

where  $\alpha \in [0, 1]$  is the target sparsity level.

For each client  $C_k$ , the overall objective of Basic-FedGR is:

$$\mathcal{L}_{rat}^k = \mathcal{L}_r^k + \lambda_{sp} \mathcal{L}_{sp}^k. \quad (3)$$

Finally, after local training, clients upload extractor ( $\Theta_k^s$ ) and predictor ( $\Theta_k^p$ ) parameters to the global server, which aggregates them via Eq. (1).

During inference, we compute predictions using the rationale representation  $\mathbf{h}_r$  that are from the global server.

## 4 Difference-based Sample Selection for Federated Graph Rationalization

While Basic-FedGR establishes a foundational approach for exploring GNN explainability in FL scenarios, its performance is fundamentally constrained by data heterogeneity among clients and the propensity for local shortcut learning. These inherent limitations diminish the framework's practical effectiveness. To overcome these challenges, in this section, we introduce a *Difference-based Sample Selection* for Federated Graph Rationalization (*DiffGR*) method.

*DiffGR* integrates three core components: (1) A sample selection strategy that identifies optimal training instances for inferring environment. (2) An environment inference method that infers latent environment information. (3) An environment-conditioned data augmentation module that maps samples from the current environment to other inferred environments.

### 4.1 Sample Selection Strategy

In federated graph rationalization tasks, current environment inference methods [19, 46] typically utilize the complement (non-rationale subgraphs) representations of samples to infer environment information through clustering algorithms. However, these methods suffer from inefficient environment inference, as the complement subgraphs may contain not only genuine environment information but also noisy components. Such noise interference during clustering can lead to imprecise environment information.

To address this limitation, we propose a difference-based sample selection strategy. Our strategy is founded on the following key assumption:

**ASSUMPTION 4.1.** *In FL scenarios, given the global server model  $f^t(\cdot)$  and the local model  $f_k^{t-1}(\cdot)$  from the previous iteration, we assume  $f^t(\cdot)$  exhibits relatively unbiased characteristics compared to  $f_k^{t-1}(\cdot)$  [39].*

This assumption leads to an important inference: when a sample shows significant prediction difference between global and local models, it indicates stronger reliance on local shortcut features, making its complement information more representative of environment information. Formally, for any sample  $G_k$  in client  $C_k$ , we define the prediction difference score:

$$\Delta_k(G_k) = |P_k(G_k) - P_g(G_k)|,$$

where  $P_k(G_k)$  and  $P_g(G_k)$  denote the prediction distributions from the local and global models, respectively. This difference score carries dual theoretical significance: (1) Higher  $\Delta_k(G_k)$  values strongly suggest the sample's dependence on client-specific shortcuts. (2) The complement components of such samples typically contain more distinctive environmental signals.

While an intuitive approach would be to select samples with  $\Delta_k(G_k) > \gamma$  using a fixed threshold  $\gamma$ , determining the optimal threshold presents substantial challenges. Instead, we propose a ranking-based selection strategy:

First, we sort all samples by their difference scores in descending order:

$$\mathcal{D}_k^{\text{sorted}} = \{G_k^1, G_k^2, \dots, G_k^{|\mathcal{D}_k|}\}, \quad (4)$$

where  $\Delta_k(G_k^1) \geq \Delta_k(G_k^2) \geq \dots \geq \Delta_k(G_k^{|\mathcal{D}_k|})$ . Then, we select the top- $n\%$  samples as potential candidates for environment inference:

$$\mathcal{D}_k^{\text{env}} = \{G_k^i \in \mathcal{D}_k^{\text{sorted}} | i \leq \lceil n\% \times |\mathcal{D}_k| \rceil\}. \quad (5)$$

## 4.2 Environment Inference

Following the sample selection phase, each client  $C_k$  processes its selected environmental samples  $\mathcal{D}_k^{\text{env}}$  to derive local environment representations. Specifically, given the rationale and complement representations  $\left\{ \left( \mathbf{h}_{r_k}^i, \mathbf{h}_{c_k}^i \right) \right\}_{i=1}^{|\mathcal{D}_k^{\text{env}}|}$ , the environment inference proceeds through:

$$E_k = \text{Clustering}(\left\{ \mathbf{h}_{c_k}^i \right\}_{i=1}^{|\mathcal{D}_k^{\text{env}}|}), \quad (6)$$

where  $E_k = \{\mathbf{e}_k^1, \mathbf{e}_k^2, \dots, \mathbf{e}_k^T\}$ , and we employ the k-means clustering algorithm [12, 19] as the  $\text{Clustering}(\cdot)$  function. Finally, the server then aggregates all client environments to obtain the global environment  $E = \{\mathbf{e}^1, \mathbf{e}^2, \dots, \mathbf{e}^{N \times T}\}$ .

## 4.3 Environment-conditioned Data Augmentation

To address the local shortcut learning problem, we propose a solution that introduces global environmental contexts to each client. Specifically, we generate counterfactual samples through Environment conditioned Data Augmentation (EcDA), effectively disrupting spurious correlations inherent in local data distributions. In this subsection, we first introduce the vanilla EcDA methods [19, 46]. Then, from the perspective of Mutual Information (MI), we explain the rationality of the design of the previously vanilla EcDA methods. Finally, based on our analysis, we further propose a new MI-inspired counterfactual data augmentation method to enhance its effectiveness.

**4.3.1 Vanilla EcDA.** Based on Basic-FedGR, given the global environment set  $E = \{\mathbf{e}^1, \mathbf{e}^2, \dots, \mathbf{e}^{N \times T}\}$  aggregated from all clients, each client  $C_k$  implements the following augmentation method. Specifically, a generator  $\mathbb{E}\mathbb{G}(\cdot)$  transforms local rationale representations  $\mathbf{h}_{r_k}^i$  according to randomly sampled environments  $\mathbf{e}^j \sim E$ :

$$\mathbf{h}_{r_k}^{i,j} = \mathbb{E}\mathbb{G}(\mathbf{h}_{r_k}^i, \mathbf{e}^j) = \mathbf{h}_{r_k}^i + \mathbf{e}^j. \quad (7)$$

Then, the predictor  $\Phi(\cdot)$  processes both original and augmented representations while maintaining label consistency (since environments should not affect true labels):

$$\hat{Y}_{r_k}^{i,j} = \Phi(\mathbf{h}_{r_k}^{i,j}), \quad \mathcal{L}_{\text{aug}}^k = \mathbb{E}_{(G_k, Y_k) \sim \mathcal{D}_k} [\ell(\hat{Y}_{r_k}^{i,j}, Y_k)]. \quad (8)$$

Finally, the objective of Vanilla EcDA is  $\mathcal{L}_{\text{rat}}^k + \lambda_{\text{aug}}^k \mathcal{L}_{\text{aug}}^k$ .

**4.3.2 A Mutual Information Perspective on Vanilla EcDA.** Before we analyze Vanilla EcDA based on MI, we first present two fundamental principles, sufficiency [43] and independence [1], which should be satisfied to compose the true rationale:

**DEFINITION 4.2. Sufficiency Principle for Rationalization:**

$$P(Y | G) = P(Y | R),$$

where the rationale  $R$  contains sufficient information from  $G$  to predict  $Y$  accurately.

**DEFINITION 4.3. Independence Principle for Rationalization:**

$$Y \perp E | R,$$

indicating that the label  $Y$  is conditionally independent of the environment  $E$  given the rationale  $R$ , where  $\perp$  denotes probabilistic independence.

Then, these principles can be formulated as a MI optimization problem [1]:

$$\max I(R; Y), \text{ s.t. } \min I(Y; E | R), \quad (9)$$

where  $I(X; Y)$  represents the MI between variables  $X$  and  $Y$ .

To implement Eq. (9), we derive the following tractable objective:

$$\begin{aligned} \mathcal{L}_{\text{mi}} &= \mathcal{L}_{\text{mi-max}} + \beta \mathcal{L}_{\text{mi-min}} \\ &= \underbrace{-\mathbb{E}[\log P(Y | R)]}_{\textcircled{1}} + \beta \underbrace{\mathbb{E}[\log P(Y | R) - \mathbb{E}_e[\log P(Y | R, e)]]}_{\textcircled{2}}. \end{aligned} \quad (10)$$

Among them, the first term  $\textcircled{1}$  is mathematically equivalent to Eq. (2). From the second term  $\textcircled{2}$ , we can observe that the ideal state is achieved when the predictive distributions of  $P(Y | R, e)$  and  $P(Y | R)$  are completely identical. In Vanilla EcDA, when both Eq. (2) and Eq. (8) reach their optimal conditions, the optimal state of the second term  $\textcircled{2}$  is also attained. Therefore, Eq. (8) can be regarded as a special case of  $\textcircled{2}$ , where  $P(Y | R, e)$  can be implemented through Eq. (7) and Eq. (8).

Below presents the proof of Eq. (10):

**PROOF. Part 1: Maximizing  $I(R; Y)$ :**

The mutual information decomposes as:

$$I(R; Y) = H(Y) - H(Y | R),$$

where  $H(Y)$  is the label entropy (constant w.r.t. optimization). Thus:

$$\max I(R; Y) \equiv \min H(Y | R).$$

By definition of conditional entropy:

$$H(Y | R) = -\mathbb{E}_{y,r} [\log P(y | r)],$$

so maximizing  $I(R; Y)$  reduces to minimizing the negative log-likelihood:

$$\max I(R; Y) \equiv \min -\mathbb{E}[\log P(Y | R)].$$

**Part 2: Minimizing  $I(Y; E | R)$ :**

The conditional mutual information expands to:

$$\begin{aligned} I(Y; E | R) &= H(Y | R) - H(Y | E, R) \\ &= -\mathbb{E}[\log P(Y | R)] + \mathbb{E}[\log P(Y | E, R)]. \end{aligned}$$

Since  $I(Y; E | R) \geq 0$  and by the inequality:

$$\begin{aligned} & -\mathbb{E}[\log P(Y | R)] + \mathbb{E}[\log P(Y | E, R)] \\ & \leq |\mathbb{E}[\log P(Y | R)] - \mathbb{E}[\log P(Y | E, R)]|, \end{aligned}$$

we obtain the upper bound:

$$\min I(Y; E | R) \equiv \min |\mathbb{E}[\log P(Y | R)] - \mathbb{E}[\log P(Y | E, R)]|.$$

#### Combined Objective:

Substituting both terms into Eq. (9) yields:

$$\begin{aligned} & \max I(R; Y) - \beta I(Y; E | R) \\ & \equiv \min -\mathbb{E}[\log P(Y | R)] + \beta |\mathbb{E}[\log P(Y | R)] - \mathbb{E}[\log P(Y | E, R)]| \\ & = \min -\mathbb{E}[\log P(Y | R)] + \beta \mathbb{E}[|\log P(Y | R) - \mathbb{E}_e[\log P(Y | R, e)]|], \end{aligned}$$

which finishes the proof.  $\square$

**4.3.3 MI-inspired EcDA.** Based on the above analysis, where Eq. (8) can be considered as a special case of ②, we adopt the second term ② to replace Eq. (8), thereby effectively constraining the validity of generated counterfactual data.

Finally, during the training, by recalling Eq. (3) and Eq. (10), the overall objective of EaDA in each client  $C_k$  is defined as :

$$\mathcal{L}_{EcDA}^k = \mathcal{L}_{rat}^k + \beta \mathcal{L}_{mi-min}^k. \quad (11)$$

In the inference phase, only  $\mathbf{h}_r$  that derived by the global server is employed to yield task results.

## 5 Experiments

In this section, we conduct comprehensive experiments to evaluate the effectiveness of *DiffGR* by investigating the following research questions (RQs):

- **RQ1:** How does *DiffGR* perform in both task prediction and rationale extraction compared to baseline methods?
- **RQ2:** What is the relative impact of different architectural components and hyperparameter choices on *DiffGR*'s performance?
- **RQ3:** To what extent does *DiffGR* maintain its performance when deployed in FL settings with varying scales?
- **RQ4:** Can *DiffGR*'s framework enhance the performance of existing rationalization methods?

### 5.1 Datasets

• **Synthetic Dataset.** In this paper, we use the Spurious-Motif dataset [37, 42] as a synthetic benchmark for motif-type prediction. Each graph in the dataset comprises two subgraphs: the motif subgraph  $R$  and the base subgraph  $C$ . The motif subgraph  $R$  determines the graph's motif type, which can be one of three classes: Cycle, House, or Crane, denoted as  $R = \{0, 1, 2\}$ . In contrast, the base subgraph  $C$  acts as a complement structure and varies depending on the motif type, with three possible forms: Tree, Ladder, or Wheel, represented as  $E = \{0, 1, 2\}$ . Thus, each graph in Spurious-Motif is labeled by its motif-base pair, such as Cycle-Tree. To introduce shortcuts into the benchmark, the dataset is constructed such that the motif subgraph is sampled uniformly, while the base subgraph is selected based on the probability distribution  $P(C) = b \times \mathbb{I}(C = R) + \frac{1-b}{2} \times \mathbb{I}(C \neq R)$ , where  $b$  controls the strength of the shortcut bias—higher values of  $b$  lead to more pronounced shortcuts. We evaluate three dataset variants with  $b = \{0.5, 0.7, 0.9\}$ . To ensure unbiased evaluation,

the test set is constructed as a de-biased (balanced) dataset by setting  $b = \frac{1}{3}$ .

**OGB.** In this paper, with a focus on real-world applications, we conduct experiments on the Open Graph Benchmark (OGB) [15]. The datasets we utilize from OGB include MolHIV, MolToxCast, MolBBBP, MolBACE, and MolSIDER. To guarantee a just and unbiased evaluation, we first utilize the default scaffold splitting approach provided in OGB. This method is employed to divide the datasets into three subsets: the training, validation and test set. It is important to note that the distributions of the test set and the training set are remarkably different, which implies that they represent distinct environments. Since the above datasets are all standard centralized datasets, we distribute the constructed training dataset to  $N$  clients by applying the unbalanced partition algorithm known as Latent Dirichlet Allocation (LDA) [13, 14]. This technique entails generating a heterogeneous partition through sampling  $p_i \sim \text{Dir}_N(\gamma)$ . For each local client, a proportion  $p_{i,n}$  of the training instances belonging to class  $i$  is allocated. In this paper, for Spurious-Motif, we set the value of  $N$  to 3 and  $\gamma$  to 3. For OGB, we configure  $N$  as 4 and  $\gamma$  as 4. A detailed description of datasets is shown in Appendix A.

### 5.2 Comparison Methods

In this section, to assess the efficacy of *DiffGR*, we initiate our evaluation by contrasting it with multiple rationale-based approaches: DisC [7], GSAT [28], CAL [33], DARE [44], and RGDA [20]. These methods have been adapted from centralized settings to FL scenarios through implementation within the Basic-FedGR framework. We refer to these adapted versions as FedDisC, FedGSAT, FedCAL, FedDARE, and FedRGDA, respectively. In addition, we conduct comparisons between *DiffGR* and two other methods: FedGR [45] and EaDA [46], both of which are federated graph rationalization techniques relying on data augmentation.

Below are the details of our comparison methods:

- **DisC** [7]: A disentanglement-based approach that learns causal and shortcut substructures in graph data. It synthesizes counterfactual training samples to decorrelate causal and shortcut variables, effectively reducing shortcut reliance.
- **GSAT** [28]: An information bottleneck-guided method that introduces stochastic masking to suppress label-irrelevant subgraphs while preserving predictive patterns, improving interpretability and robustness.
- **CAL** [33]: Causal Attention Learning (CAL) framework that constructs causal rationales and reduces shortcut confounding effects through attention mechanisms, enhancing model generalization.
- **DARE** [44]: A self-supervised disentanglement method that extracts comprehensive input representations and rationale features without explicit supervision.
- **RGDA** [20]: Generates counterfactual samples using bias substructures but lacks explicit disentanglement mechanisms to fully isolate bias features from original inputs.
- **FedGR** [45]: Federated learning approach that creates shortcut-conflicted samples via difference-aware augmentation, operating under client environment-agnostic conditions.
- **EaDA** [46]: Environment-aware data augmentation that exploits task-irrelevant complement subgraphs to infer environmental

**Table 1: Performance on the Synthetic and OGB Dataset with the GIN backbone.**

|                       | Spurious-Motif (ACC)   |                        |                        | OGB (AUC)              |                        |                        |                        |                        |
|-----------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                       | bias=0.5               | bias=0.7               | bias=0.9               | MolHIV                 | MolToxCast             | MolBBBP                | MolSIDER               | MolBACE                |
| Basic-FedGR           | 0.3182 ± 0.0353        | 0.3681 ± 0.0359        | 0.3031 ± 0.0291        | 0.6985 ± 0.0155        | 0.6111 ± 0.0055        | 0.6339 ± 0.0142        | 0.5774 ± 0.0175        | 0.7058 ± 0.0334        |
| FedDisC               | 0.4418 ± 0.0182        | 0.4481 ± 0.0381        | 0.3579 ± 0.0471        | 0.7212 ± 0.0201        | 0.6274 ± 0.0018        | 0.6561 ± 0.0121        | 0.5869 ± 0.0142        | 0.7253 ± 0.0290        |
| FedCAL                | 0.4213 ± 0.0109        | 0.5289 ± 0.0087        | 0.4191 ± 0.0248        | 0.7039 ± 0.0113        | 0.6170 ± 0.0051        | 0.6575 ± 0.0076        | 0.5879 ± 0.0138        | 0.7248 ± 0.0212        |
| FedGSAT               | 0.4281 ± 0.0328        | 0.5259 ± 0.0381        | 0.4194 ± 0.0338        | 0.7149 ± 0.0226        | 0.6255 ± 0.0030        | 0.6555 ± 0.0085        | 0.5952 ± 0.0082        | 0.7369 ± 0.0413        |
| FedDARE               | 0.4483 ± 0.0193        | 0.4891 ± 0.0391        | 0.4288 ± 0.0977        | 0.7220 ± 0.0165        | 0.6289 ± 0.0059        | 0.6621 ± 0.0096        | 0.5886 ± 0.0113        | 0.7301 ± 0.0092        |
| FedRGDA               | 0.4087 ± 0.0293        | 0.5089 ± 0.0198        | 0.4286 ± 0.0313        | 0.7246 ± 0.0085        | 0.6235 ± 0.0034        | 0.6605 ± 0.0157        | 0.5906 ± 0.0151        | 0.7282 ± 0.0301        |
| FedGR                 | 0.4610 ± 0.0289        | 0.5538 ± 0.0398        | 0.4977 ± 0.0315        | 0.7387 ± 0.0186        | 0.6316 ± 0.0054        | 0.6690 ± 0.0174        | 0.6017 ± 0.0202        | 0.7435 ± 0.0170        |
| EaDA                  | 0.5269 ± 0.0273        | 0.5892 ± 0.0163        | 0.5447 ± 0.0365        | 0.7611 ± 0.0084        | 0.6345 ± 0.0108        | 0.6713 ± 0.0077        | 0.6178 ± 0.0040        | 0.7743 ± 0.0073        |
| <i>DiffGR</i>         | <b>0.5558 ± 0.1540</b> | <b>0.6693 ± 0.0530</b> | <b>0.5560 ± 0.0579</b> | <b>0.7653 ± 0.0092</b> | <b>0.6534 ± 0.0021</b> | <b>0.6794 ± 0.0054</b> | <b>0.6291 ± 0.0064</b> | <b>0.7908 ± 0.0355</b> |
| <i>DiffGR</i> -sample | 0.5348 ± 0.0983        | 0.6239 ± 0.0376        | 0.5489 ± 0.0326        | 0.7629 ± 0.0076        | 0.6387 ± 0.0232        | 0.6733 ± 0.0077        | 0.6198 ± 0.0058        | 0.7823 ± 0.0128        |

**Table 2: Performance on the Synthetic and OGB Dataset with the GCN backbone.**

|                       | Spurious-Motif (ACC)   |                        |                        | OGB (AUC)              |                        |                        |                        |                        |
|-----------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                       | bias=0.5               | bias=0.7               | bias=0.9               | MolHIV                 | MolToxCast             | MolBBBP                | MolSIDER               | MolBACE                |
| Basic-FedGR           | 0.3219 ± 0.0401        | 0.3589 ± 0.0292        | 0.3024 ± 0.0487        | 0.6998 ± 0.0111        | 0.6043 ± 0.0149        | 0.6393 ± 0.0080        | 0.5718 ± 0.0108        | 0.7715 ± 0.0104        |
| FedDisC               | 0.4369 ± 0.0486        | 0.4584 ± 0.0378        | 0.3673 ± 0.0931        | 0.7342 ± 0.0186        | 0.6171 ± 0.0094        | 0.6406 ± 0.0039        | 0.5894 ± 0.0076        | 0.7782 ± 0.0081        |
| FedCAL                | 0.4438 ± 0.0477        | 0.5173 ± 0.0462        | 0.4284 ± 0.0832        | 0.7485 ± 0.0127        | 0.6205 ± 0.0046        | 0.6524 ± 0.0257        | 0.5957 ± 0.0116        | 0.7843 ± 0.0198        |
| FedGSAT               | 0.4394 ± 0.0915        | 0.5383 ± 0.0326        | 0.4398 ± 0.0534        | 0.7457 ± 0.0079        | 0.6125 ± 0.0032        | 0.6536 ± 0.0085        | 0.6004 ± 0.0246        | 0.7887 ± 0.0138        |
| FedDARE               | 0.4472 ± 0.0471        | 0.4782 ± 0.0474        | 0.4327 ± 0.0372        | 0.7424 ± 0.0368        | 0.6094 ± 0.0089        | 0.6416 ± 0.0159        | 0.5968 ± 0.0292        | 0.7928 ± 0.0144        |
| FedRGDA               | 0.4187 ± 0.0375        | 0.4987 ± 0.0744        | 0.4377 ± 0.0432        | 0.7293 ± 0.0166        | 0.6197 ± 0.0049        | 0.6456 ± 0.0061        | 0.5958 ± 0.0143        | 0.7496 ± 0.0136        |
| FedGR                 | 0.4580 ± 0.0531        | 0.5526 ± 0.0624        | 0.4918 ± 0.0619        | 0.7571 ± 0.0104        | 0.6282 ± 0.0092        | 0.6693 ± 0.0149        | 0.6093 ± 0.0039        | 0.7881 ± 0.0246        |
| EaDA                  | 0.5082 ± 0.0471        | 0.5431 ± 0.0577        | 0.5273 ± 0.0662        | 0.7354 ± 0.0179        | 0.6463 ± 0.0040        | 0.6516 ± 0.0103        | 0.6157 ± 0.0154        | 0.7649 ± 0.0195        |
| <i>DiffGR</i>         | <b>0.5323 ± 0.0617</b> | <b>0.5793 ± 0.0642</b> | <b>0.5604 ± 0.0161</b> | <b>0.7651 ± 0.0118</b> | <b>0.6531 ± 0.0050</b> | <b>0.6603 ± 0.0156</b> | <b>0.6227 ± 0.0093</b> | <b>0.8038 ± 0.0110</b> |
| <i>DiffGR</i> -sample | 0.5134 ± 0.0487        | 0.5502 ± 0.0378        | 0.5330 ± 0.0218        | 0.7588 ± 0.00157       | 0.6491 ± 0.0083        | 0.6579 ± 0.0117        | 0.6198 ± 0.0094        | 0.7901 ± 0.0145        |

contexts and generate augmented samples, improving generalization.

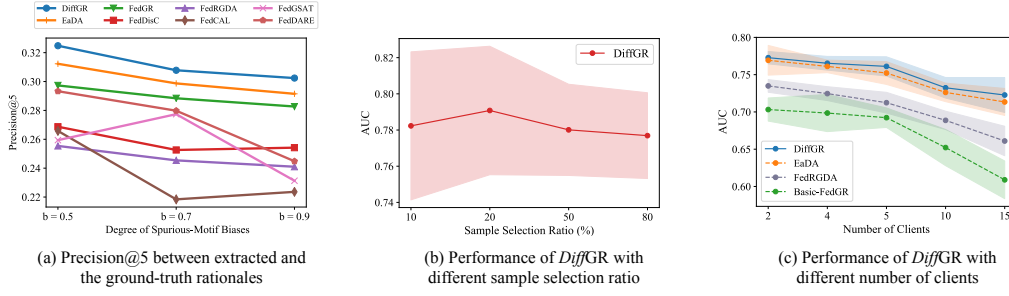
### 5.3 Experimental Setup

In the evaluation process, the AUC metric is employed to evaluate the task prediction performance of datasets from OGB, whereas the ACC metric is utilized for Spurious-Motif. Given that Spurious-Motif provides ground-truth rationales, we are able to evaluate the accuracy of the extracted rationales using the Precision@5 metric. This metric determines the correctness of the top 5 extracted rationales by comparing them against the ground-truth rationales. The hyperparameters  $\lambda_{sp}$  is fixed at 1.0 and  $\beta$  is 0.1 across all experiments. The hidden dimensionality  $d$  is set to 128 for the OGB datasets and 32 for the Spurious-Motif dataset. For optimization, we use the Adam optimizer [17] with an initial learning rate of  $1 \times 10^{-3}$  for OGB and  $1 \times 10^{-2}$  for Spurious-Motif. The number of inferred environments per client varies by dataset: 5 for MolHIV and MolToxCast, 10 for the remaining OGB datasets, and 3 for Spurious-Motif. In the sample selection strategy (Eq. (5)), we

set  $n = 20$ . Following [45], the predefined sparsity  $\alpha$  is configured as 0.1 for MolHIV, 0.5 for MolSIDER, MolToxCast, MolBBBP, and MolBACE, and 0.4 for all other datasets. Training proceeds over 20 communication rounds ( $T_c$ ), with 10 epochs per round, resulting in a total of 200 iterations. For model implementation, we adopt GIN and GCN as the backbone architectures for both our proposed methods and baseline comparisons. All methods are trained on a single A100 GPU with 5 distinct random seeds.

### 5.4 Overall Performance (RQ1)

**Performance of the Task Prediction.** To evaluate the effectiveness of *DiffGR*, we first conduct a comparative analysis of various baseline methods in terms of task prediction. The experimental results are presented in Table 1 and Table 2. Specifically, it can be observed from the table that for baseline methods migrated from centralized scenarios (e.g., FedCAL and FedRGDA), their performance is decent and shows significant improvement compared Basic-FedGR. However, their performance is not as good as FedGR and EaDA, both of which are rationalization methods specifically



**Figure 1: (a) Results of Precision@5 between extracted rationales and the ground-truth rationales on Spurious-Motif. (b) Varying the specified sample selection ratio on MolBACE. (c) Performance of *DiffGR* with different number of clients on MolBACE.**

designed for FL scenarios. This indicates that rationalization methods from centralized scenarios cannot be directly applied to FL scenarios. Instead, such methods need to be designed with the characteristics of FL in mind. Moreover, compared to EADA, which clusters the complement representations of all samples as environment information, *DiffGR* outperforms on all datasets, demonstrating the effectiveness of our difference-based sample selection strategy in sampling optimal training instances to infer environment information.

**Performance of the Rationale Extraction.** Additionally, to assess whether *DiffGR* avoids generating rationales by relying on shortcut features, we perform experiments on the Spurious-Motif dataset, which contains ground-truth rationales. As shown in Figure 1(a), our method consistently surpasses the baseline approaches across varying levels of bias in the Spurious-Motif dataset. This demonstrates that *DiffGR* effectively mitigates data heterogeneity challenges and reduces the impact of local shortcut learning, thereby improving the reliability of rationale extraction.

## 5.5 Ablation Study (RQ2)

In this section, we validate the effectiveness of each module in *DiffGR* framework through ablation experiments in Table 1 and Table 2. First, we assess the importance of the sampling module by constructing a variant called *DiffGR*-sample. This variant removes the difference-based sampling strategy and instead directly uses the complement information of all samples for environment inference. The experimental results show that *DiffGR*-sample underperforms compared to the full *DiffGR*, demonstrating the critical role of our proposed sampling strategy in selecting optimal training instances for inferring environmental information. Next, we further examine the contribution of the MI-inspired EcDA module. When this module is removed, *DiffGR* reduces to Basic-FedGR. The experimental results reveal a significant performance gap between Basic-FedGR and *DiffGR*, highlighting the necessity of EcDA after obtaining environment representations. Specifically, the MI-inspired EcDA module effectively leverages environmental information to guide the data augmentation process, thereby enhancing the model’s generalization capability.

## 5.6 Sensitivity Analysis (RQ2)

***DiffGR* with Different Dataset Partitioning Methods.** In this subsection, we first examine how *DiffGR* performs under different

federated dataset partitioning strategies. To this end, we intentionally introduce more imbalanced data distributions in experiments. For the binary classification datasets MolHIV, MolBBBP, and MolBACE (where labels are either 0 or 1), we deliberately create class imbalances across clients. The exact partitioning scheme is detailed in Table 7. As shown in the table, the repartitioning results in significant variations in the positive-to-negative class ratios among clients. For the multi-label classification datasets, MolToxCast and MolSIDER that contain multiple classes, we partition the data based on graph size, assigning graphs with fewer nodes to one client and those with more nodes to another. Further details on this partitioning approach are provided in Table 7. An analysis of the table reveals notable disparities in node/edge counts across graphs from different clients, confirming the presence of imbalanced data distributions. We then evaluate *DiffGR* and baseline methods using these repartitioned datasets. The results presented in Table 3 demonstrate that *DiffGR* consistently outperforms other methods, even under varying dataset partitioning schemes. This experiment underscores the generalizability of our approach, proving its effectiveness across diverse FL scenarios with differing data distribution characteristics.

***DiffGR* with Different Sample Selection Ratio.** To systematically evaluate the impact of the sample selection ratio  $n$  on model performance, we designed multiple comparative experiments ( $n \in \{10, 20, 50, 80\}$ ). As shown in Figure 1(b), the model achieves the optimal balance when  $n=20$ . Notably, an excessively small sampling ratio ( $n=10$ ) may fail to capture the diversity of client data distributions, resulting in the loss of some genuinely useful environmental signals and consequently leading to performance degradation. Conversely, an excessively large ratio ( $n=80$ ) may introduce excessive noise, compromising the representation of environmental information. This experiment fully demonstrates the necessity of the sampling based environmental inference.

## 5.7 Scalability Analysis (RQ3)

The ability to scale efficiently is a critical requirement in FL scenarios, where the number of participating clients can vary significantly. To thoroughly examine the scalability of *DiffGR*, we conduct experiments by progressively increasing the number of clients and observing the corresponding changes in model performance. As illustrated in Figure 1(c), while the overall performance of all rationalization-based methods tends to decrease as the number of clients grows, *DiffGR* consistently demonstrates superior robustness compared to

**Table 3: Performance on the Real-world Dataset in FL scenarios with another partition method.**

|               | MolHIV                                | MolToxCast                            | MolBBBP                               | MolSIDER                              | MolBACE                               |
|---------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| Basic-FedGR   | 0.6877 $\pm$ 0.0180                   | 0.5943 $\pm$ 0.0042                   | 0.6232 $\pm$ 0.0118                   | 0.5313 $\pm$ 0.0118                   | 0.6832 $\pm$ 0.0248                   |
| FedDisC       | 0.7102 $\pm$ 0.0031                   | 0.6082 $\pm$ 0.0031                   | 0.6438 $\pm$ 0.0048                   | 0.5423 $\pm$ 0.0193                   | 0.7088 $\pm$ 0.0234                   |
| FedCAL        | 0.6987 $\pm$ 0.0130                   | 0.5985 $\pm$ 0.0058                   | 0.6489 $\pm$ 0.0032                   | 0.5489 $\pm$ 0.0024                   | 0.7123 $\pm$ 0.0387                   |
| FedGSAT       | 0.7083 $\pm$ 0.0034                   | 0.6055 $\pm$ 0.0046                   | 0.6518 $\pm$ 0.0024                   | 0.5573 $\pm$ 0.0137                   | 0.7177 $\pm$ 0.0303                   |
| FedDARE       | 0.6829 $\pm$ 0.0177                   | 0.6021 $\pm$ 0.0049                   | 0.6482 $\pm$ 0.0083                   | 0.5498 $\pm$ 0.0294                   | 0.7003 $\pm$ 0.0205                   |
| FedRGDA       | 0.7031 $\pm$ 0.0035                   | 0.5953 $\pm$ 0.0060                   | 0.6502 $\pm$ 0.0095                   | 0.5512 $\pm$ 0.0078                   | 0.7276 $\pm$ 0.0320                   |
| FedGR         | 0.7290 $\pm$ 0.0061                   | 0.6179 $\pm$ 0.0159                   | 0.6654 $\pm$ 0.0121                   | 0.5697 $\pm$ 0.0028                   | <b>0.7743 <math>\pm</math> 0.0145</b> |
| EaDA          | 0.7308 $\pm$ 0.0243                   | 0.6206 $\pm$ 0.0104                   | 0.6748 $\pm$ 0.0184                   | 0.5735 $\pm$ 0.0039                   | 0.7636 $\pm$ 0.0093                   |
| <i>DiffGR</i> | <b>0.7337 <math>\pm</math> 0.0058</b> | <b>0.6276 <math>\pm</math> 0.0085</b> | <b>0.6798 <math>\pm</math> 0.0033</b> | <b>0.6148 <math>\pm</math> 0.0171</b> | 0.7696 $\pm$ 0.0058                   |

**Table 4: Structural Generalizability of *DiffGR*. Each method with *DiffGR* is highlighted in gray.**

|                | MolHIV                     | MolToxCast                 | MolBBBP                    | MolSIDER                   | MolBACE                    |
|----------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| FedGSAT        | 0.7149                     | 0.6255                     | 0.6555                     | 0.5952                     | 0.7369                     |
| <i>+DiffGR</i> | 0.7528 ( $\uparrow$ 3.79%) | 0.6334 ( $\uparrow$ 0.79%) | 0.6728 ( $\uparrow$ 1.73%) | 0.6029 ( $\uparrow$ 0.77%) | 0.7584 ( $\uparrow$ 2.15%) |
| FedDARE        | 0.7220                     | 0.6289                     | 0.6621                     | 0.5886                     | 0.7301                     |
| <i>+DiffGR</i> | 0.7587 ( $\uparrow$ 3.67%) | 0.6382 ( $\uparrow$ 0.93%) | 0.6749 ( $\uparrow$ 1.28%) | 0.5984 ( $\uparrow$ 0.98%) | 0.7663 ( $\uparrow$ 3.62%) |
| FedRGDA        | 0.7246                     | 0.6235                     | 0.6605                     | 0.5906                     | 0.7282                     |
| <i>+DiffGR</i> | 0.7688 ( $\uparrow$ 4.42%) | 0.6379 ( $\uparrow$ 1.44%) | 0.6745 ( $\uparrow$ 1.40%) | 0.6033 ( $\uparrow$ 1.27%) | 0.7547 ( $\uparrow$ 2.65%) |

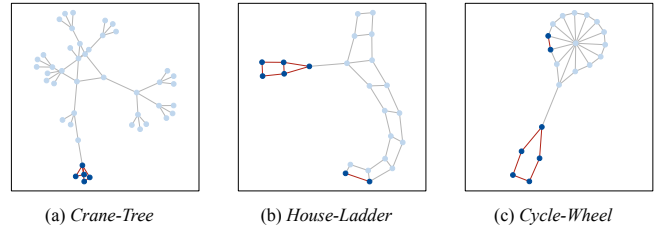
the baselines. Notably, even under more challenging settings with a larger client population, *DiffGR* is able to maintain a relatively high AUC, indicating its enhanced capacity to generalize across distributed environments. For example, as the number of clients increases from  $N = 4$  to  $N = 15$ , the AUC of *DiffGR* decreases by only 4.26%, in contrast to a significantly larger drop of 8.96% observed for Basic-FedGR. This substantial difference highlights the effectiveness of *DiffGR* in preserving performance under scale, showcasing its strong scalability and adaptability in real-world FL deployments where client heterogeneity and distribution shift are often unavoidable.

### 5.8 Structural Generalizability of *DiffGR* (RQ4)

From Eq. (11), we observe that *DiffGR* introduces an additional loss function ( $\mathcal{L}_{mi-min}^k$ ) compared to Basic FedGR. This observation naturally leads to an important research question: *Can DiffGR enhance the performance of other graph rationalization methods in FL settings?* To investigate this, we integrate *DiffGR* with three approaches: FedDARE, FedRGDA, and FedGR, and conduct comprehensive evaluations. As demonstrated in Table 4, the incorporation of *DiffGR* consistently improves performance across all baseline methods. These results strongly suggest that *DiffGR*'s framework demonstrates significant generalizability, being compatible with diverse rationalization architectures.

### 5.9 Case Study

We evaluate *DiffGR*'s rationale extraction capability on the Spurious-Motif dataset (bias level  $b = 0.9$ ). As illustrated in Figure 2, each input graph consists of a motif (e.g., Cycle, House, or Crane) attached to a base structure (e.g., Tree, Wheel, or Ladder). The rationale nodes identified by *DiffGR* are highlighted in navy blue,

**Figure 2: Visualization of *DiffGR* rationale subgraphs.**

where a node is classified as part of the rationales if its predicted probability  $\hat{m}_i$  exceeds 0.55. Additionally, edges connecting two rationale nodes are marked with red lines. The qualitative results clearly demonstrate that *DiffGR* is capable of identifying compact and semantically meaningful substructures that align well with the ground-truth motifs. These findings demonstrate that *DiffGR* successfully extracts high-quality rationales, significantly improving both the interpretability and predictive performance of the model.

## 6 Conclusion

In this paper, we proposed *DiffGR*, a difference-based sample selection strategy for federated graph rationalization, to address the challenge of shortcut learning caused by data heterogeneity in FL scenarios. By identifying samples where local and global models exhibit the highest disagreement, *DiffGR* improved the quality of environment inference and enable more robust cross-environment data augmentation. Furthermore, we introduced a MI-inspired data augmentation framework to enhance the extraction of environment-invariant rationales by minimizing spurious correlations while preserving causal predictive features. Experiments on real-world and synthetic datasets demonstrate the effectiveness of *DiffGR* in improving federated graph rationalization performance.

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**Table 5: Statistics of Spurious-Motif Datasets. Among them, different clients share the same valid and test set.**

|                                  | b=0.5                          | b=0.7                         | b=0.9                          |
|----------------------------------|--------------------------------|-------------------------------|--------------------------------|
| Client1/Client2/Client3/Val/Test | 377/662/1961/3,000/6,000       | 377/662/1,961/3,000/6,000     | 377/662/1,961/3,000/6,000      |
| Classes                          | 3                              | 3                             | 3                              |
| Avg. Nodes                       | 18.60/18.29/18.48/18.50/88.80  | 18.73/18.27/18.8/18.50/88.80  | 19.02/18.54/18.66/18.50/88.80  |
| Avg. Edges                       | 27.72/27.31/27.55/27.54/125.14 | 28.29/27.3/28.05/27.54/125.14 | 28.74/27.63/27.81/27.54/125.14 |

**Table 6: Statistics of OGB Datasets.**

|                                          | MolHIV                               | MolToxCast                          | MolBACE                             |
|------------------------------------------|--------------------------------------|-------------------------------------|-------------------------------------|
| Client1/Client2/Client3/Client4/Val/Test | 9,380/6,148/10,113/7,260/4,113/4,113 | 871/614/3,819/1,556/858/858         | 425/234/191/360/151/152             |
| Classes                                  | 2                                    | 617                                 | 2                                   |
| Avg. Nodes                               | 25.31/25.32/25.15/25.27/27.79/25.27  | 16.41/16.86/16.63/16.91/26.17/28.19 | 33.81/33.91/33.28/33.33/37.23/34.82 |
| Avg. Edges                               | 54.19/54.2/53.89/54.15/61.05/55.59   | 32.91/33.93/33.45/33.99/56.09/60.71 | 73.06/73.13/71.87/72.09/81.3/75.11  |

|                                          | MolBBBP                             | MolSIDER                             |
|------------------------------------------|-------------------------------------|--------------------------------------|
| Client1/Client2/Client3/Client4/Val/Test | 472/299/325/535/204/204             | 422/333/201/185/143/143              |
| Classes                                  | 2                                   | 27                                   |
| Avg. Nodes                               | 22.44/22.15/22.34/22.81/33.20/27.51 | 28.85/30.96/30.97/29.7/43.24/53.27   |
| Avg. Edges                               | 48.42/47.53/48.05/49.19/71.84/59.75 | 60.53/64.77/64.87/62.25/91.85/112.66 |

**Table 7: Statistics of OGB Datasets with an another partitioning method.**

|                                                      | MolHIV                         | MolBBBP                       | MolBACE                       |
|------------------------------------------------------|--------------------------------|-------------------------------|-------------------------------|
| Client1/Client2/Client3/Val/Test                     | 1,000/30,500/1,401/4,113/4,113 | 1,062/200/369/204/204         | 900/150/160/151/152           |
| Class Ratio (Positive(1): Negative(0)) among clients | 500:500/500:30,000/232:1,169   | 1,000:62/100:100/269:100      | 300:600/100:50/80:80          |
| Avg. Nodes                                           | 25.02/25.03/30.34/27.79/25.27  | 21.56/23.36/24.69/33.20/27.51 | 34.18/34.12/29.86/37.23/34.82 |
| Avg. Edges                                           | 52.80/53.61/65.60/61.05/55.59  | 46.43/50.03/53.33/71.84/59.75 | 73.83/73.61/64.74/81.3/75.11  |

|                                  | MolToxCast                    | MolSIDER                        |
|----------------------------------|-------------------------------|---------------------------------|
| Client1/Client2/Client3/Val/Test | 3,000/2,000/1,860/858/858     | 400/500/241/143/143             |
| Classes                          | 617                           | 27                              |
| Avg. Nodes                       | 9.42/16.32/28.79/26.17/28.19  | 12.66/25.53/67.95/43.24/53.27   |
| Avg. Edges                       | 17.73/32.92/59.75/56.09/60.71 | 24.96/54.86/142.12/91.85/112.66 |

## A Data Statistics

To evaluate the effectiveness of *DiffGR*, we conduct experiments on multiple datasets. For synthetic data, we use the Spurious-Motif dataset [37, 42]. For real-world evaluation, we employ the Open Graph Benchmark (OGB) [15], specifically the OGB-Mol subsets

(MolHIV, MolToxCast, MolBBBP, MolBACE, and MolSIDER), which provide diverse molecular properties for analysis. Dataset statistics are summarized in Table 5 and Table 6. Additionally, Table 7 shows per-client data distributions after partitioning OGB using alternative methods.